1. Core KPI Query

```
SELECT
 ROUND(SUM(totalrevenue_cleaned), 2) AS totalrevenue_cleaned,
 COUNT(DISTINCT invoiceno) AS total_orders,
 COUNT(DISTINCT customerid) AS total_customers,
 SUM(quantity) AS total_quantity,
 ROUND(SUM(totalrevenue_cleaned) / COUNT(DISTINCT invoiceno), 2) AS avg_order_value,
 ROUND(SUM(totalrevenue_cleaned) / COUNT(DISTINCT customerid), 2) AS revenue_per_customer
FROM `big-query-practice-465600.E_commerce_portfolio.supply_chain_kpi`
```

Query results

+ Create conversation Save results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	totalrevenue_clea...	total_orders	total_customers	total_quantity	avg_order_value	revenue_per_cust...
1	7093089.62	18127	4303	3789928	391.3	1648.41

2. Order - Level Metrics

```
SELECT
 invoiceno,
 SUM(totalrevenue_cleaned) AS order_revenue,
 SUM(quantity) AS total_items
FROM `big-query-practice-465600.E_commerce_portfolio.supply_chain_kpi`
GROUP BY invoiceno
ORDER BY order_revenue DESC
```

Query results

Job information | Results | Visualization | JSON | Executi

Row	invoiceno	order_revenue	total_items
1	556444	38970.0	60
2	556917	12572.33	5913
3	572035	11991.91999999...	4948
4	578305	10584.77000000...	2111
5	563614	10494.44	4132
6	552978	8951.26	3682
7	580727	8935.980000000...	2193
8	560580	8801.509999999...	2565
9	576339	8425.540000000...	2175
10	569650	8404.489999999...	2971
11	545475	8171.879999999...	2928

3. Country - Level Metrics

```
SELECT
 country_cleaned,
 ROUND(SUM(totalrevenue_cleaned), 2) AS total_revenue,
 COUNT(DISTINCT invoiceno) AS total_orders,
 COUNT(DISTINCT customerid) AS total_customers,
 ROUND(SUM(totalrevenue_cleaned) / COUNT(DISTINCT invoiceno), 2) AS aov,
 ROUND(SUM(totalrevenue_cleaned) / COUNT(DISTINCT customerid), 2)
   AS revenue_per_customer
FROM `big-query-practice-465600.E_commerce_portfolio.supply_chain_kpi`
GROUP BY country_cleaned
ORDER BY total_revenue DESC
```

Query results

+ Create conversation Save results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	country_cleaned	total_revenue	total_orders	total_customers	aov	revenue_per_cust...
1	United Kingdom	5821304.99	16268	3887	357.84	1497.63
2	Germany	221202.62	457	94	484.03	2353.22
3	Eire	206675.76	257	3	804.19	68891.92
4	France	198182.65	389	87	509.47	2277.96
5	Netherlands	157952.92	85	9	1858.27	17550.32
6	Australia	72916.77	51	9	1429.74	8101.86
7	Switzerland	55754.19	51	21	1093.22	2654.96
8	Spain	50840.54	89	30	571.24	1694.68
9	Belgium	40841.46	98	25	416.75	1633.66
10	Norway	34833.83	36	10	967.61	3483.38
11	Portugal	33015.84	57	19	579.23	1737.68
12	Japan	22986.37	16	8	1436.65	2873.3

Row	country_cleaned	total_revenue	total_orders	total_customers	aov	revenue_per_cust...
13	Finland	21943.92	41	12	535.22	1828.66
14	Channel Islands	18743.99	25	8	749.76	2343.0
15	Denmark	17744.94	18	9	985.83	1971.66
16	Italy	17283.08	38	14	454.82	1234.51
17	Sweden	16386.87	33	8	496.57	2048.36
18	Singapore	16217.33	7	1	2316.76	16217.33
19	Cyprus	13223.49	16	8	826.47	1652.94
20	Austria	9403.8	17	11	553.16	854.89
21	Poland	7334.65	19	6	386.03	1222.44
22	Israel	7215.84	5	3	1443.17	2405.28
23	Greece	4760.52	5	4	952.1	1190.13
24	Iceland	4060.4	7	1	580.06	4060.4

Row	country_cleaned	total_revenue	total_orders	total_customers	aov	revenue_per_cust...
25	Usa	3580.39	5	4	716.08	895.1
26	Canada	3071.66	5	3	614.33	1023.89
27	Malta	2725.59	5	2	545.12	1362.8
28	Unspecified	2660.77	8	4	332.6	665.19
29	United Arab Emirates	1902.28	3	2	634.09	951.14
30	Lebanon	1693.88	1	1	1693.88	1693.88
31	Lithuania	1661.06	4	1	415.26	1661.06
32	European Community	1300.25	4	1	325.06	1300.25
33	Brazil	1143.6	1	1	1143.6	1143.6
34	Rsa	1002.31	1	1	1002.31	1002.31
35	Czech Republic	826.74	2	1	413.37	826.74
36	Bahrain	548.4	2	2	274.2	274.2
37	Saudi Arabia	145.92	1	1	145.92	145.92

4. Monthly Trends

```
SELECT
 year,
 month,
 ROUND(SUM(totalrevenue_cleaned), 2) AS monthly_revenue,
 COUNT(DISTINCT invoiceno) AS total_orders
FROM `big-query-practice-465600.E_commerce_portfolio.supply_chain_kpi`
GROUP BY year, month
ORDER BY year, month
```

Job information | **Results** | Visualization | JSON | Execution details

Row	year	month	monthly_revenue	total_orders
1	2010	12	467994.4	1361
2	2011	1	394795.7	968
3	2011	2	378004.41	971
4	2011	3	492245.83	1296
5	2011	4	392480.0	1118
6	2011	5	576436.92	1518
7	2011	6	532293.85	1366
8	2011	7	486549.2	1305
9	2011	8	506738.83	1246
10	2011	9	760383.03	1720
11	2011	10	825494.81	1890
12	2011	11	1002385.92	2610
13	2011	12	277286.72	758

5. Repeat Customer Analysis

```
WITH customer_orders AS (
 SELECT
   customerid,
   COUNT(DISTINCT invoiceno) AS num_orders
 FROM `big-query-practice-465600.E_commerce_portfolio.supply_chain_kpi`
```

```
  GROUP BY customerid
)

SELECT
  COUNTIF(num_orders = 1) AS one_time_customers,
  COUNTIF(num_orders > 1) AS repeat_customers,
  ROUND(COUNTIF(num_orders > 1) / COUNT(*), 4) AS repeat_rate
FROM customer_orders
```

Row	one_time_custo...	repeat_customers	repeat_rate
1	1486	2817	0.6547

6. Top Customers

```
SELECT
  customerid,
  ROUND(SUM(totalrevenue_cleaned), 2) AS total_spent
FROM `big-query-practice-465600.E_commerce_portfolio.supply_chain_kpi`
GROUP BY customerid
ORDER BY total_spent DESC
LIMIT 10
```

Job information | Results | Visualizatic

Row	customerid	total_spent
1	14646	152712.6
2	14911	132910.61
3	17511	71615.64
4	14156	69423.94
5	14096	65155.75
6	12415	60801.17
7	15311	56232.71
8	13089	55890.16
9	18102	54656.02
10	14088	50491.81

7. Data Validation

```
SELECT
 ROUND(SUM(totalrevenue_cleaned), 2) AS stored_revenue,
 ROUND(SUM(quantity * unitprice_cleaned), 2) AS recalculated_revenue,
 ROUND(SUM(totalrevenue_cleaned) - SUM(quantity * unitprice_cleaned), 5)
AS difference
FROM `big-query-practice-465600.E_commerce_portfolio.supply_chain_kpi`
```

Job information | Results | Visualization | JSON

Row	stored_revenue	recalculated_reve...	difference
1	7093089.62	7093089.62	0.0

8. Invalid Values Check

```
SELECT *
```

```
FROM `big-query-practice-465600.E_commerce_portfolio.supply_chain_kpi`
WHERE quantity <= 0 OR unitprice_cleaned <= 0
```

★ **No Data, indicating there are no invalid values in this dataset**

9. Consistency Check

```
SELECT
 COUNT(*) AS total_rows,
 COUNT(DISTINCT invoiceno) AS unique_orders,
 COUNT(DISTINCT customerid) AS unique_customers
FROM `big-query-practice-465600.E_commerce_portfolio.supply_chain_kpi`
```

Job information | **Results** | Visualization | JSON

Row	total_rows	unique_orders	unique_customers
1	388180	18127	4303